

Complete Git, GitHub & Batch Automation Notes

1. The Standard GitHub Workflow

To take a project from a local folder to a live GitHub repository, execute this exact sequence:

```
git init -> git add . -> git commit -m "Initial commit" -> git  
branch -M main -> git remote add origin URL -> git push -u origin  
main
```

2. Common Git Commands

Repository Setup & Information

- `git init` : Initializes a new Git repository.
- `git clone URL` : Downloads a repository from a remote server.
- `git status` : Shows the current state of the working directory and staging area.
- `git log --oneline` : Displays a compact history of commits.
- `git diff` : Shows the exact lines of code that have been changed.

Staging & Committing

- `git add .` : Stages all modified and untracked files in the current directory.
- `git commit -m "message"` : Saves the staged changes permanently with a descriptive message.

Branching & Merging

- `git branch` : Lists all local branches.
- `git checkout branch` : Switches to the specified branch.
- `git switch branch` : A newer command to switch branches.
- `git merge branch` : Combines the specified branch into your current active branch.

Remote Syncing

- `git pull origin main` : Fetches and merges changes from the remote main branch to your local branch.
- `git push origin main` : Uploads your local main branch commits to the remote repository.
- `git fetch` : Downloads history from the remote repository without merging it.

Modifying & Undoing Changes

- `git stash / git stash pop` : Temporarily shelves changes and then reapplies them later.
- `git reset --soft HEAD~1` : Undoes the last commit but keeps your files and staging area intact.
- `git reset --hard HEAD~1` : Completely undoes the last commit and permanently deletes any uncommitted changes.
- `git restore file` : Discards unstaged changes in a specific file.
- `git rm file` : Removes a file from your working directory and staging area.
- `git mv old new` : Renames a file or moves it to a new location.

3. Remote Commands

These commands are used specifically for managing the connections between your local repository and external servers (like GitHub):

- `git remote -v` : Lists all currently configured remote repositories and their URLs.
- `git remote add origin https://github.com/user/repo.git` : Connects your local repository to a specific URL.
- `git remote remove origin` : Disconnects the remote repository named 'origin'.
- `git remote set-url origin NEW_URL` : Changes the URL of an existing remote connection.

4. Windows Batch Scripting (.bat) Automation

Common .bat Commands

You can utilize the following commands for automation:

```
echo , @echo off , cls , pause , title , color , set VAR=value , if , goto , call ,  
start , cd , dir , mkdir , rmdir , del , copy , move , ren , xcopy , robocopy ,  
tasklist , taskkill , ping , ipconfig , shutdown , exit .
```

Example BAT Script for Git Automation

This script automates the process of adding, committing, and pushing code:

```
@echo off  
echo Building...  
git add .  
git commit -m "Auto commit"  
git push origin main  
pause
```